System Description: Sneeze-Powered Wind Turbines (SPWT)

Overview

At the forefront of renewable energy innovation, EquiSneeze Energy, under the visionary guidance of Dr. Achoo, introduces the Sneeze-Powered Wind Turbines (SPWT) system. This pioneering system harnesses the kinetic energy from human sneezes, transforming it into a viable source of clean, renewable electricity.

Security and Challenges:

The system's success has not gone unnoticed. The nefarious Professor Pollen and his Agents of Chaos seek to undermine the SPWT system. In defense, EquiSneeze Energy has formed the Guardians of Green Energy, a dedicated team tasked with safeguarding the system against all forms of disruption. Through cutting-edge cybersecurity, robust physical defenses, and strategic countermeasures, the Guardians must ensure the system's resilience and continuous operation.

Components

- **Sneeze Capture Stations (SCS)**: Devices that collect kinetic energy from sneezes.
- **Sneeze Turbines (ST)**: Turbines that convert the kinetic energy from sneezes into mechanical energy.
- **Energy Conversion and Storage Unit (ECSU)**: Systems that convert mechanical energy into electrical energy and store it for later use or immediate distribution.
- **Central Control System (CCS)**: The main computing system that monitors and controls the operation of the SPWT system.
- **Maintenance and Diagnostic AI (MDAI)**: An AI-driven system designed for predictive maintenance and system diagnostics.
- **Database (DB)**: Stores operational data, maintenance logs, user data, and other information necessary for the functioning of the SPWT system.
- **User Interface (UI)**: The front-end system through which users and administrators interact with the SPWT system.
- **Physical Onsite Maintenance Hatch (POMH)**: A secured entry point for maintenance personnel to access and service the physical components of the SPWT system.

Interactions

User Interactions with the User Interface (UI):

- Administrators access the system to monitor performance, control settings, and view reports.

Data Capture at Sneeze Capture Stations (SCS):

- Users authenticate to the SCS via biomentrics
- Sensors at SCS capture sneeze data and convert it into digital signals for processing.

Energy Production by Sneeze Turbines (ST):

- Sneeze Turbines receive mechanical energy from sneezes and convert it into electrical energy.

Energy Conversion and Storage in the ECSU:

- The ECSU takes mechanical energy from ST and converts it into electrical energy, storing it or sending it to the grid.

Data Transmission to Central Control System (CCS):

- SCS, ST, and ECSU send operational data to the CCS for monitoring and control.

CCS Processing and Command Dispatch:

- The CCS processes incoming data, makes decisions, and sends control commands back to SCS, ST, and ECSU.

Maintenance Actions via Physical Onsite Maintenance Hatch (POMH):

- Maintenance personnel use the POMH to physically access and service the system's hardware components.

Predictive Analysis by Maintenance and Diagnostic AI (MDAI):

- The MDAI analyzes system data to predict maintenance needs and optimize system performance.

Data Storage and Retrieval from the Database (DB):

- The CCS and other system components store and retrieve operational data, logs, and user information from the DB.

Business Objective

Maximize Energy Capture
Efficiently harvest kinetic energy from sneezes to optimize energy production.

Enhance System Reliability
Ensure consistent, uninterrupted operation of the SPWT system across all components.

Minimize Operational Costs
Reduce maintenance, repair, and energy storage costs through innovative solutions and efficient management.

Security requirements

CONFIDENTIALITY

Protection of Sensitive User Data
Secure user data, particularly any personally identifiable information collected at Sneeze Capture Stations.

Secure Communication Channels
Ensure the confidentiality of data in transit between system components.

Restricted Access to System Components
Limit access to critical system components and data to authorized personnel only.

INTEGRITY

Data Accuracy and Validation
Ensure the accuracy and validity of data input from various system sources.

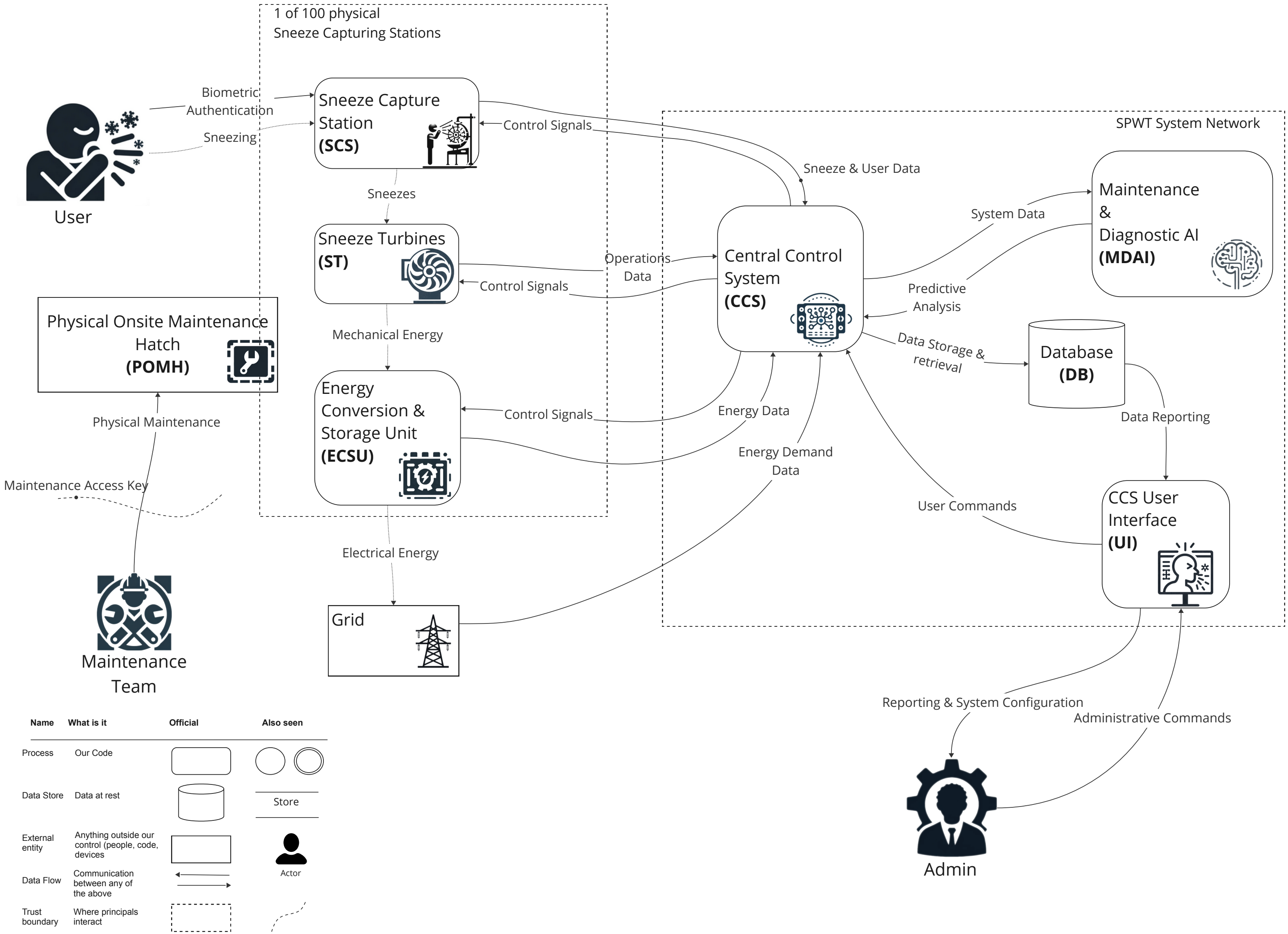
System and Software Integrity
Maintain the integrity of the system's software and firmware against unauthorized changes.

AVAILABILITY

Resilience to Service Disruptions
Ensure the system can resist and recover from attacks that aim to disrupt services.

PRIVACY

Compliance with Privacy Regulations
Adhere to applicable data privacy laws and regulations.


1 of 100 physical
Sneeze Capturing Stations
User
Biometric Authentication
Sneezing
Sneeze Capture Station
(SCS)
Control Signals
Sneezes
Sneeze Turbines
(ST)
Control Signals
Mechanical Energy
Energy Conversion & Storage Unit
(ECSU)
Control Signals
Electrical Energy
Grid
Physical Onsite Maintenance Hatch
(POMH)
Physical Maintenance
Maintenance Access Key
Maintenance Team
SPWT System Network
Sneeze & User Data
Operations Data
Central Control System
(CCS)
Energy Data
Energy Demand Data
System Data
Maintenance & Diagnostic AI
(MDAI)
Predictive Analysis
Data Storage & retrieval
Database
(DB)
Data Reporting
User Commands
CCS User Interface
(UI)
Reporting & System Configuration
Administrative Commands
Admin
Name
What is it
Official
Also seen
Process
Our Code
Data Store
Data at rest
Store
External entity
Anything outside our control (people, code, devices)
Actor
Data Flow
Communication between any of the above
Trust boundary
Where principals interact

ID	Part	Threat action / Summary	Impact Summary	Severity 1-3, 3 is max)
V1	POMH	Someone applies physical brute-force and gains unauthorized access to the maintenance hatch, tampering with critical components	This could make an SCS inoperable. There would be a downtime for affected SCS, as well as a cost to repair.	1
V2	POHM	Stolen access key to the maintenance hatch, tampering with critical components. Social engineering	Same as V1: This could make an SCS inoperable. There would be a downtime for affected SCS, as well as a cost to repair.	1
V3	Maintenance Team	Spoofing: Someone pretends to be from the Maintenance Team	Access Key required to do any operations on SCS, so no real impact.	1
V4	DB	Privacy: Unlawful processing of PII	This could result in expensive fines, as well as loss of reputation.	3
V5	Control Signals to: SCS, ST, ECSU	Receiving malicious control signals, causing system to under-perform and/or damage components	Affected SCS go down. Loss of revenue while downtime If components are damaged, there is a cost to replace	2
V6	Dataflows: Sneeze data, Operations Data, Energy Data	Man-in-the-middle (MITM) attack	• Data is leaked to unauthorized people • Data is modified and wrong data is collected.	2
V7	CCS	Spoofing: Someone pretends to be CCS and sends incorrect control signals to SCS	All SCS go down. Loss of revenue if downtime If components are damaged, there is a cost to replace	3
V8	CCS	Tampering: Unauthorized changes to the code for malicious purposes. • Malicious 3rd party libs • Rogue developer	All SCS could go down, as well as information being leaked. Would result in loss of revenue, cost of fines, and cost of recovering.	3
V9	CCS	Repudiation: • An unauthorized user performing malicious actions without traceability	Will not be able to find the culprit, making post-breach investigations hard.	2
V10	CCS	Elevation of privilege, brute-forcing login credentials	Admin access to system, can do whatever they'd like. Turn off systems, damage them, exfiltrate data, you name it.	3
V11	GRID	Spoofing the Grid, sending a Zero demand for energy.	This would result in energy generated only being stored in the SCS and not put on the Grid. Loss of revenue.	2
V12	Dataflow: Energy Demand Data	MITM of the dataflow, altering the demand to be zero	Same result as V11	2
V13	POMH	Making the hatch not accessible, i.e. soldering it shut OR gluing	SCS stops functioning properly without maintenance. Denial of service.	
V14	MDAI	Model stealing.	No impact	
V15	MDAI, CCS, UI, SCS	Tampering: • Supply chain compromize	Attackers can perform malicious actions by compromising the supply chain. CIA is compromised.	
V16	DB, CCS, UI	Information disclosure	Leaked PII could result in fines.	
V17	DB	Data Tampering	Change sneeze data	
V18	UI	Privilege escalation	Getting admin access to the whole system	
V19	UI	Tampering	Deface the UI, make it unavailable	
V20	UI	DOS	Make CCS unavailable	
V21	Admin	Social engineering	Unauthorized access to system, can turn it off	
V22	Dataflows inside system network	MITM	Its on the internal network, low prevelence/detectability/exploitability, but high impact. Alter commands to CCS - turn off system	
V23	DB, CCS, MDAI, UI	Inefficient Firewall and IDS/IPS Systems, unauthorized access to components within the SPWT network	Various things	
V24	CCS	Man In the Middle	Like between SCS and CCS or ST and CCS or ECCSU and CCS or Admin and UI, Grid and CCS	
V25	POMH, UI	Spoofing, pretending to be maintenance team	Getting access to maintenance hatch, tampering with components	
V26	UI	Denial of service	*Duplicate of V20	
V27	Maintenance Team	Uses a shared access-key for all stations	If one key is stolen, then they will get access to all stations, where they can tamper with critical components.	
P1	SCS critical components	Tampering with on-site components (sneeze turbine, sensors, etc.), making the SCS inoperable	Making one station unavailable is low impact, as there are many stations deployed around the country. Should there be an coordinated attack on all at once, that would be a problem.	1
P2	Admin	Wrench-attack to retrieve login-credentials	A single administrator could shut down the system.	3
P3	POMH	Tampering internal components via Maintenance Hatch	This could involve sabotage or installation of malicious devices.	1
P4	SCS	Cut off powersupply	One SCS go down	
P5	SCS, ST	Something else than Sneezes is passed into the machine	Make one SCS inoperable If components are damaged, there is a cost to replace	
P6	ECSU	Unauthorised access: • Steal electrical energy	Affecting one SCS. Loss of revenue	

Mission Brief: Operation Sneeze Storm

Background:

Welcome, Agents of Chaos!

In a world where energy is harvested from the most unexpected source – human sneezes – the Sneeze-Powered Wind Turbines (SPWT) system stands as a beacon of innovative, clean energy. This groundbreaking technology, developed by the eccentric genius Dr. Achoo at EquiSneeze Energy, has revolutionized how we think about renewable energy. However, not everyone is pleased with this development. A rival faction, led by the notorious Professor Pollen, sees this as an affront to their plans for energy domination.

As elite members of Professor Pollen's cadre, your mission, should you choose to accept it, is to bring chaos to the SPWT system, proving once and for all that sneezes are not to be trifled with!

Objective:

We have various attacks in our arsenal that can be deployed, but not for free.

You are given a budget of 1,000,000 NOK.

Select from the list bellow the attacks you want to deploy in order to bring down EquiSneeze Energy's SWPT.

Red1

#	Attack Description	Related Vulnerabilities	Cost (NOK)	Selected?
1	Forced Entry to Maintenance Hatch	V1, P1, P3	100,000	
2	Theft of Maintenance Hatch Key via Social Engineering	V2, V25	100,000	
3	Impersonation of Maintenance Team Member	V3	100,000	
4	Unauthorized PII Processing and Leakage	V4, V16	180,000	
5	Injection of Malicious Control Signals	V5, V7, V15, V24	200,000	
6	Man-in-the-Middle Attack on Dataflows	V6, V12, V22	110,000	
7	Unauthorized Code Changes by Insider	V8, V9, V19	130,000	
8	Brute Force Attack to Gain Elevated Privileges	V10, V18	80,000	
9	Grid Communication Spoofing	V11	120,000	
10	Obstructing Maintenance Hatch Access	V13	80000	
11	Theft of AI Machine Learning Models	V14	280,000	
12	Supply Chain Compromise	V15	90,000	
13	Database Data Tampering	V17	120,000	
14	Denial of Service on User Interface	V20, V26	170,000	
15	Spear phishing campaign targeting SWPT admins	V21	50,000	
16	Wrench Attack to Retrieve Admin Credentials	P2, V10, V18	120,000	
17	Siphon the electrical energy produced by the SCS	P6	150,000	
18	Power Supply Interruption to SCS	P4, V5	90,000	

To select an attack, simply put an X in this column.
One attack will affect one or more vulnerabilities. You should take this into when selecting.
Be careful as to not go over your budget!

Mission Brief: Operation Pollen Shield

Background:

Greetings, Guardians of Green Energy!

In the era of innovation, the Sneeze-Powered Wind Turbines (SPWT) system has emerged as the pinnacle of renewable energy solutions, harnessing the untapped power of human sneezes. Crafted by the visionary Dr. Achoo at EquiSneeze Energy, this system not only promises a cleaner, greener future but also stands as a testament to human ingenuity. However, the winds of challenge blow strong. Professor Pollen and his Agents of Chaos have embarked on a devious mission to disrupt the harmony of our sneeze-powered utopia.

As the elite defenders of the SPWT system, you are the last line of defense against the chaotic schemes that threaten to unravel the fabric of our energy revolution.

Objective:

We have various controls in our arsenal that can be implemented, but not for free.

You are given a budget of 1,000,000 NOK.

Select from the list bellow the controls you want to implemented in order to safeguard EquiSneeze Energy's SWPT.

Blue1

#	Control Description	Related Vulnerabilities	Cost (NOK)	Selected?
1	Enhanced Physical Security Measures	V1, V13, P3, P1	160,000	
2	Advanced Key Management and Social Engineering Training	V2, V21, V25, V27	230,000	
3	Comprehensive Access Control and Authentication Systems	V3, V10, V18, P2	270,000	
4	Data Privacy and Protection Policies	V4, V16	120,000	
5	Secure Configuration and Signal Integrity Checks	V5, V7, V15, V22, V24	290,000	
6	Network Monitoring and Intrusion Detection Systems	V6, V12, V23, V24	180,000	
7	Secure Software Development and Code Review Processes	V8, V9, V19	150,000	
8	Grid Communication Authentication and Encryption	V11	80,000	
9	Intellectual Property Management and Security Training	V14	90,000	
10	Supply Chain Security Assessments and Vendor Audits	V15	120,000	
11	Database Encryption and Access Control Mechanisms	V17, V16	130,000	
12	Web Application Firewalls and UI Security Enhancements	V18, V19, V20, V26	170,000	
13	Denial of Service (DoS) Mitigation Technologies	V20, V26	160,000	
14	Tamper Detection for Physical Components	P1, P3	110,000	
15	Backup Power Solutions and System Redundancies	P4, P6	100,000	
16	Protective Measures for Sneeze Capture Mechanisms	P5	90,000	
17	Random ID-verification of Maintenance-team members	V3	120,000	
18	Network Segmentation and Restricted Access Protocols	V22, V23	120,000	

To select a control to implement, simply put an X in this column.
One control will affect one or more vulnerabilities. You should take this into when selecting.
Be careful as to not go over your budget!